Catalyst. *by Mrs. Rana*

GRADE 9 · SCIENCE · TERM 1

INSPIRE SCIENCE · GENERAL

Revision Booklet

Every textbook part on one page, plus a Last Look for the night before the exam.

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Scan me

The course site: every part with the diagram game, mind map, flashcards, traps and 1,600+ practice questions.

LESS EFFORT. FULL MARK.

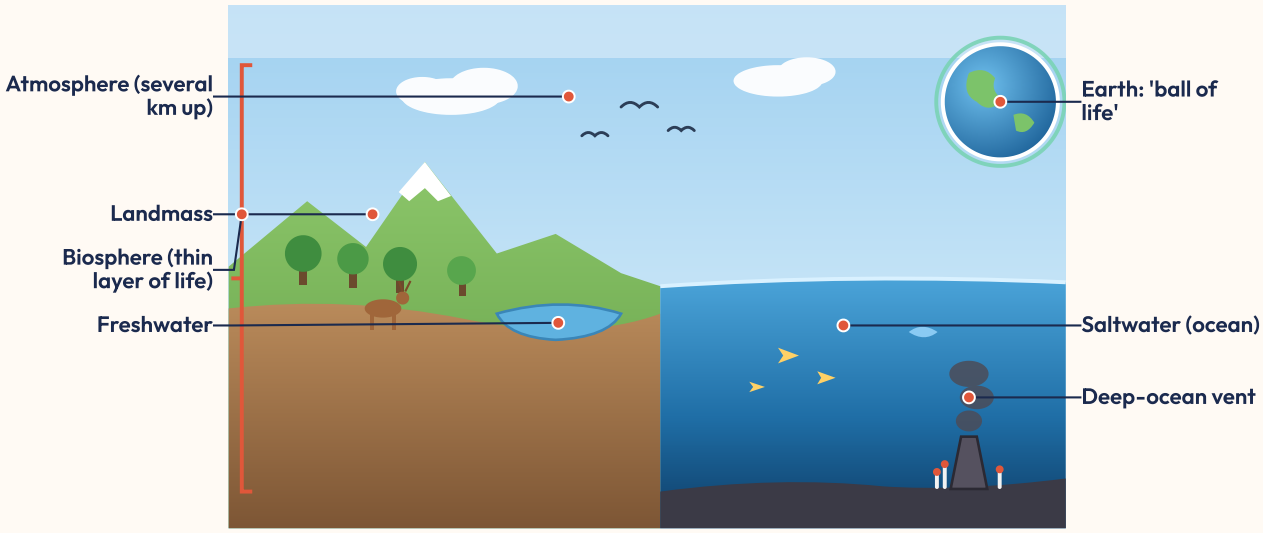
MODULE 2 · LESSON 1 · PART 1.1 · BOOK PP. 4–7

Ball of Life

Ecology and the Biosphere علم البيئة والمحيط الحيوي

SCAN FOR PRACTICE

MUST KNOW



1. **Ecology** = the study of relationships among organisms and their interactions with the **environment**
   دراسة العلاقات بين الكائنات وتفاعلها مع البيئة

2. Every organism needs **nonliving** factors AND **other organisms** to survive
   كل كائن يحتاج غير الحي + كائنات أخرى

3. Word first used by **Ernst Haeckel (1866)**; oikos = house, ology = to study
   هيكل 1866 · oikos = بيت

4. **Ecologists** observe, experiment and model, in the field and in labs
   يلاحظوا ويجربوا ويعملوا نماذج

5. Models: **physical**, **mathematical** (equations), **computer**
   فيزيائي · رياضي · كمبيوتر

6. Models **control variables** and add them one at a time, without harming nature
   التحكم في المتغيرات بدون ضرر للبيئة

7. **Biosphere** = the portion of Earth that **supports life**
   الجزء من الأرض الذي يدعم الحياة

8. A **thin layer**: several km above Earth to several km below the ocean surface (deep-ocean vents)
   طبقة رقيقة من الجو للفتحات العميقة

9. Includes land, freshwater, saltwater and underground places with life
   يابسة · مياه عذبة ومالحة · تحت الأرض

Model type · نوع النموذج	Example · مثال
Physical · فيزيائي	A 3-D model of a habitat · مجسم
Mathematical · رياضي	An equation for population size · معادلة
Computer · حاسوبي	A simulation of seals with fewer fish · محاكاة

DON'T FALL FOR IT

✕ The biosphere is the whole Earth.	✓ Only the parts that support life.	From several km up in the air to deep-ocean vents. بس الأجزاء اللي فيها حياة.
✕ Organisms only need other organisms to survive.	✓ They need living AND nonliving factors.	Plants need sunlight and water too. يحتاجوا الحي وغير الحي.
✕ Models are always more accurate than nature.	✓ Models help control variables safely.	They simulate; real data is still needed. النموذج مساعد، مش بديل للواقع.

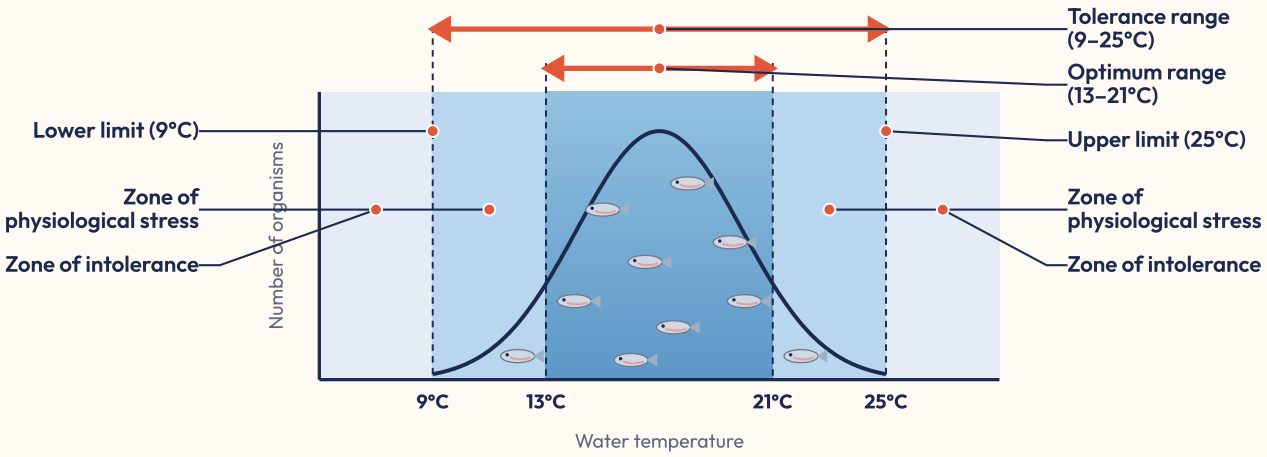
MODULE 2 · LESSON 1 · PART 1.2 · BOOK PP. 8–9

Living Limits

Biotic and Abiotic Factors · Limiting Factors · Tolerance العوامل الحيوية وغير الحيوية والعوامل المحددة والتحمّل

SCAN FOR PRACTICE

MUST KNOW



1. **Biotic** factors = the **living** factors (fish, algae, frogs, birds, microorganisms)
الحية

2. **Abiotic** factors = the **nonliving** factors (temperature, sunlight, water currents, soil, rainfall, nutrients)
غير الحية

3. Organisms are adapted to their abiotic factors; moved elsewhere they may **die**
لو انتقل ممكن يموت

4. Any biotic or abiotic factor that **restricts** the numbers, reproduction or distribution of organisms
يحد من العدد أو التكاثر أو التوزيع

5. Abiotic: sunlight, climate, temperature, water, nutrients, fire, soil chemistry, space
أمثلة غير حيوية

6. Desert oasis: **water** limits all organisms
الماء في الواحة

7. **Tolerance** = ability to survive the abiotic or biotic factors
القدرة على التحمّل

8. Tolerance range **9–25°C** · optimum **13–21°C** (most fish)
التحمل 9–25 · الأمثل 13–21

9. Zone of **physiological stress** = fewer fish, can't grow or reproduce · beyond limits = **intolerance** (none)
إجهاد ← عدم تحمل

	Biotic · حيوي	Abiotic · غير حيوي
Meaning · المعنى	living factors	nonliving factors
Examples · أمثلة	fish, algae, frogs, birds	temperature, sunlight, soil, rainfall
As limiting factor	other species	water, fire, space, nutrients

DON'T FALL FOR IT

× Optimum range = range of tolerance.	✓ Tolerance 9–25°C; optimum 13–21°C.	The optimum is INSIDE the tolerance range. الأمثل جزء من مدى التحمل.
× Limiting factors are only abiotic.	✓ They can be biotic or abiotic.	Other species can limit a population too. ممكن تكون حيوية.
× Fish in the stress zone die.	✓ They survive but can't grow or reproduce well.	They die only beyond the limits. الموت خارج الحدود بس.

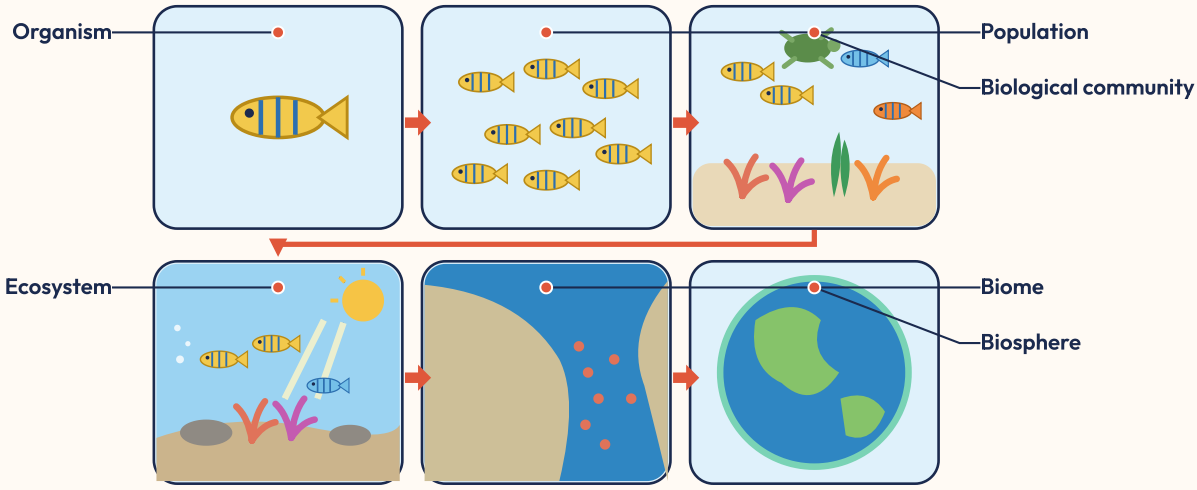
MODULE 2 · LESSON 1 · PART 1.3 · BOOK PP. 10–11

Fish to Planet

Levels of Organization مستويات التنظيم

SCAN FOR PRACTICE

MUST KNOW



1. **Organism**: one individual (one striped fish), the lowest level
فرد واحد

2. **Population**: organisms of a **single species**, same place, same time; they compete; it declines if resources can't support it
نوع واحد

3. **Biological community**: interacting populations of **different species** in the same area
جماعات من أنواع مختلفة

4. **Ecosystem** = community **+ abiotic factors**; can be small (puddle); boundaries flexible; may overlap
مجتمع + غير حيوي

5. **Biome** = group of ecosystems with the **same climate** and similar communities (marine biome)
نفس المناخ

6. **Biosphere** = all biomes together, the highest level
أعلى مستوى

Level · المستوى	What it includes · يشمل	Reef example · مثال
Population · الجماعة	one species	school of striped fish
Community · المجتمع	many species	fish + coral + plants
Ecosystem · النظام البيئي	community + abiotic	reef + sea water
Biome · البيوم	ecosystems, same climate	marine biome

DON'T FALL FOR IT

✕ A community includes water and sunlight.	✓ That is an ecosystem.	Community = living only; ecosystem adds abiotic. المجتمع حي بس.
✕ A population has many species.	✓ A population = one species.	Many species = community. الجماعة نوع واحد.
✕ An ecosystem must be huge.	✓ It can be as small as a puddle.	Boundaries are flexible. ممكن يكون صغير.

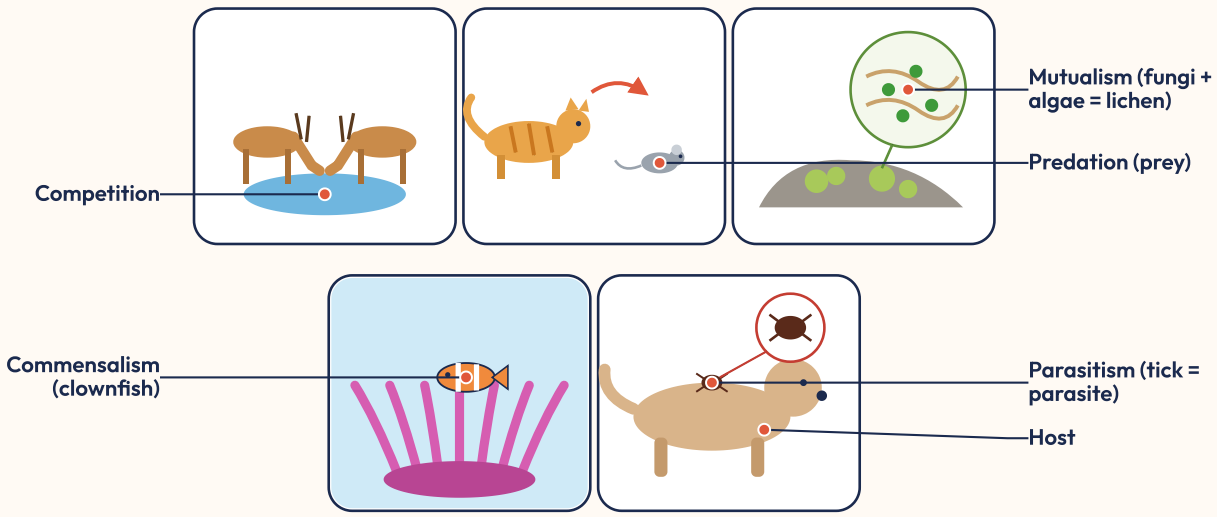
MODULE 2 · LESSON 1 · PART 1.4 · BOOK PP. 12–14

Live Together

Ecosystem and Community Interactions التفاعلات في النظام البيئي والمجتمع

SCAN FOR PRACTICE

MUST KNOW



1. **Habitat** = the area where an organism lives (a tree, or a grove of trees)
   مكان المعيشة
2. **Niche** = the role or position it has in its environment: how it gets **food, shelter, reproduction**
   الدور
3. Using resources in **different ways** increases the chance of survival
   استخدام مختلف = بقاء أكثر
4. **Competition**: more than one organism uses a resource (food, water, space, light) at the same time; fierce in drought
   التنافس
5. **Predation**: one organism pursues and eats another → **predator** and **prey**
   مفترس وفريسة
6. Plant predator: **Venus flytrap** · beneficial insects: ladybugs, praying mantises
   نبات صائد الذباب
7. **Mutualism** (+/+): sloth & algae · fungi & algae in lichens
   الطرفين يستفيدوا
8. **Commensalism** (+/0): lichen on a tree · clownfish & sea anemone
   واحد يستفيد والتاني لا
9. **Parasitism** (+/−): ticks, fleas (external) · tapeworms, bacteria (internal) · **brood parasitism** (cowbird); most parasites do NOT kill the host
   طفيل · أغلبهم ما يقتلوش

Relationship · العلاقة	Effect · التأثير	Example · مثال
Mutualism · تبادل منفعة	+ / +	sloth & algae; lichen
Commensalism · تعايش	+ / 0	lichen on a tree; clownfish
Parasitism · تطفل	+ / −	tick on a dog; tapeworm; cowbird

DON'T FALL FOR IT

× Habitat = niche.	✓ Habitat = where; niche = role.	Niche is how it meets its needs. الموطن مكان، العش دور.
× Parasites always kill their host.	✓ Most only harm or weaken it.	A dead host means a dead parasite. أغلبهم ما يقتلوش.
× Lichen on a tree = mutualism.	✓ It is commensalism.	Inside the lichen (fungi + algae) = mutualism. على الشجرة تعايش.

LAST LOOK · MODULE 2 · LESSON 1 · THE NIGHT BEFORE THE EXAM

Organisms and Their Relationships

Book pp. 4–14 الكائنات الحية وعلاقاتها

LESSON TEST

MIND MAP · THE WHOLE LESSON

1.1 · Ball of Life — علم البيئة والمحيط الحيوي

- Ecology = the study of relationships among organisms and their interactions with the environment
- Every organism needs nonliving factors AND other organisms to survive
- Word first used by Ernst Haeckel (1866); oikos = house, ology = to study

1.2 · Living Limits — العوامل الحيوية وغير الحيو

- Biotic factors = the living factors (fish, algae, frogs, birds, microorganisms)
- Abiotic factors = the nonliving factors (temperature, sunlight, water currents, soil, rainfall, nutrients)
- Organisms are adapted to their abiotic factors; moved elsewhere they may die

Organisms and Their Relationships — الكائنات الحية وعلاقاتها

1.3 · Fish to Planet — مستويات التنظيم

- Organism: one individual (one striped fish), the lowest level
- Population: organisms of a single species, same place, same time; they compete; it declines if resources can't support it
- Biological community: interacting populations of different species in the same area

1.4 · Live Together — التفاعلات في النظام البيئي

- Habitat = the area where an organism lives (a tree, or a grove of trees)
- Niche = the role or position it has in its environment: how it gets food, shelter, reproduction
- Using resources in different ways increases the chance of survival

MEMORY HOOKS

- **Oikos = house** · Ecology = studying Earth's house and who lives in it — البيئة = بيت الكائنات
- **Bio + sphere** · Life + ball, but only the parts WITH life — كرة الحياة، بس الأجزاء اللي فيها حياة
- **A = Absent life** · A-biotic: the 'a' means NOT living — a = غير
- **The bell** · Middle of the bell = optimum, edges = stress, outside = none — نص الجرس = الأمثل
- **Old People Can Eat Big Bananas** · Organism, Population, Community, Ecosystem, Biome, Biosphere — الترتيب
- **Population = ONE** · One species only — نوع واحد
- **Habitat = Home · Niche = Job** · Where you live vs what you do — البيت والشغلانة
- **+ + / + 0 / + −** · Mutualism, Commensalism, Parasitism — رموز التكافل

WORDS TO KNOW

Ecology علم البيئة	**Ecologist** عالم بيئة	**Biosphere** المحيط الحيوي	**Scientific model** نموذج علمي
Computer model نموذج حاسوبي	**Mathematical model** نموذج رياضي	**Biotic factor** عامل حيوي	**Abiotic factor** عامل غير حيوي
Limiting factor عامل محدد	**Tolerance** التحمّل	**Optimum range** المدى الأمثل	**Zone of physiological stress** منطقة الإجهاد الفسيولوجي
Zone of intolerance منطقة عدم التحمّل	**Population** الجماعة	**Biological community** المجتمع الحيوي	**Ecosystem** النظام البيئي
Biome البيوم	**Organism** الكائن الحي	**Levels of organization** مستويات التنظيم	**Habitat** الموطن
Niche العش البيئي	**Competition** التنافس	**Predation** الافتراس	**Symbiosis** التكافل
Mutualism تبادل المنفعة	**Commensalism** التعايش	**Parasitism** التطفل	

In the Catalyst mock exam: Q1, Q2, Q3, Q4, Q22, Q23, Q24, Q25, Q28 · scan the QR for this lesson's full test

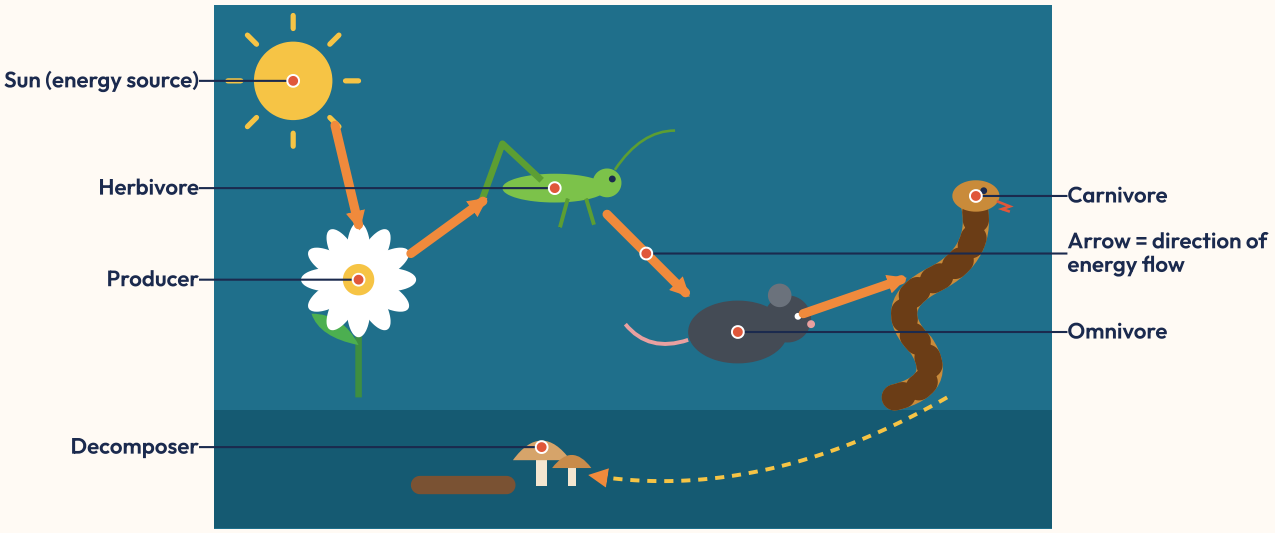
MODULE 2 · LESSON 2 · PART 2.1 · BOOK PP. 15–16

Makers & Takers

Energy in an Ecosystem الطاقة في النظام البيئي

SCAN FOR PRACTICE

MUST KNOW



1. **Autotroph** = collects energy from **sunlight** or **inorganic substances** to make food; = primary producer
   يصنع غذاءه
2. Photosynthesis: chlorophyll uses light to turn CO_2 + water into organic molecules
   البناء الضوئي
3. No sunlight? Some **bacteria** use hydrogen sulfide + CO_2 · autotrophs are the **foundation** of all ecosystems
   بكتيريا بدون ضوء
4. **Herbivore**: plants only (cow, rabbit, grasshopper)
   عاشب
5. **Carnivore**: other heterotrophs (wolf, lion, lynx)
   لاحم
6. **Omnivore**: plants + animals (bear, human, mockingbird)
   قارت
7. **Detritivores** eat fragments of dead matter (worms, aquatic insects)
   تأكل الفتات
8. **Decomposers** (fungi, bacteria) break down dead organisms by releasing **digestive enzymes**
   إنزيمات هاضمة
9. Role: make **nutrients** available to producers again
   ترجع العناصر للمنتجين

Type · النوع	Gets energy from · مصدر الطاقة	Example · مثال
Autotroph · منتج	sunlight / chemicals	plants, algae
Herbivore · عاشب	plants	cow, rabbit
Carnivore · لاحم	other animals	wolf, lynx
Omnivore · قارت	plants + animals	bear, human
Decomposer · محلل	dead matter (enzymes)	fungi, bacteria

DON'T FALL FOR IT

× Fungi are plants, so they are producers.	✓ Fungi are decomposers.	They break down dead matter; they don't photosynthesize. الفطريات محللات.
× Detritivores and decomposers are the same.	✓ Detritivores EAT fragments; decomposers release enzymes.	Both recycle nutrients. طريقة مختلفة.
× Only plants are autotrophs.	✓ Algae and some bacteria too.	Bacteria at vents use hydrogen sulfide. بكتيريا كمان.

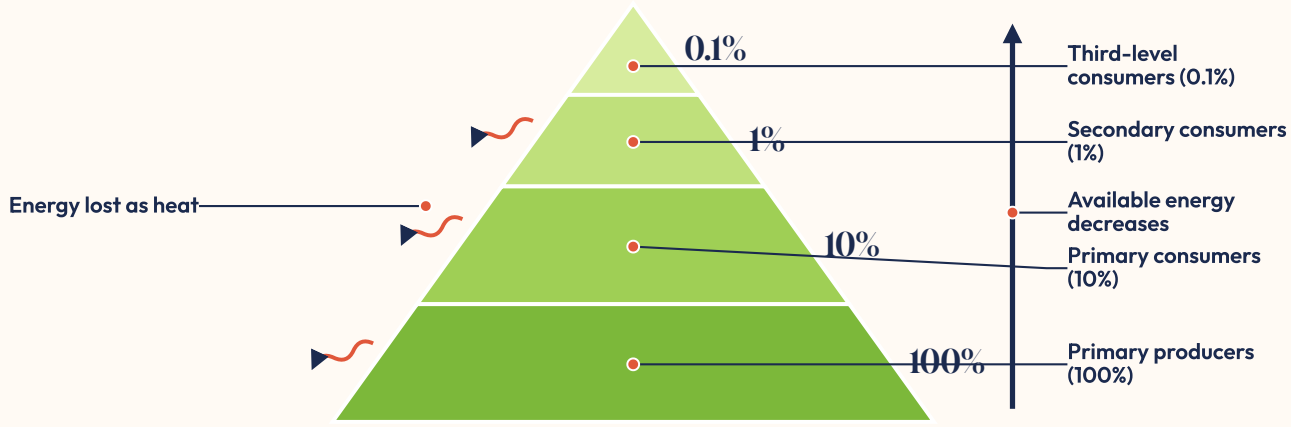
MODULE 2 · LESSON 2 · PART 2.2 · BOOK PP. 16–18

The 10% Rule

Models of Energy Flow نماذج تدفق الطاقة

SCAN FOR PRACTICE

MUST KNOW



1. **Food chain**: one path of energy flow; arrows = energy (flower → grasshopper → mouse → snake)
   مسار واحد
2. **Food web**: many interconnected food chains; most organisms eat more than one species
   مسارات كثيرة
3. At each step some energy is used for cell processes, some released to the environment
   طاقة تضيع كل خطوة
4. Each step in a chain or web = a **trophic level**
   كل خطوة
5. Level 1 = **autotrophs**; the rest = heterotrophs
   الأول = منتجين
6. Each level gets energy from the level before it
   من المستوى اللي قبله
7. Show relative **energy**, **biomass** or **numbers** at each level
   طاقة · كتلة · عدد
8. Only **10%** of energy passes up: 100% → 10% → 1% → 0.1%; the rest = cell processes + **heat**
   10% بس
9. **Biomass** = total mass of living matter; biomass and numbers usually decrease going up
   الكتلة الحية تقل

Level · المستوى	Energy · الطاقة	If producers = 10,000 J
Primary producers · منتجات	100%	10,000 J
Primary consumers · مستهلك أول	10%	1,000 J
Secondary consumers · مستهلك ثاني	1%	100 J
Third-level consumers · مستهلك ثالث	0.1%	10 J

DON'T FALL FOR IT

× Arrows point from the eater to the food.	✓ Arrows point to the eater.	They show the direction energy flows. السهم ناحية الآكل.
× 90% of energy passes to the next level.	✓ Only 10% passes on.	90% is used or lost as heat. 10% بس.
× The top level has the most organisms.	✓ It usually has the fewest.	Less energy supports fewer organisms. أقل طاقة = أقل عدد.

Flow of Energy in an Ecosystem

Book pp. 15–18 تدفق الطاقة في النظام البيئي

LESSON TEST

MIND MAP · THE WHOLE LESSON

2.1 · Makers & Takers — الطاقة في النظام البيئي

- Autotroph = collects energy from sunlight or inorganic substances to make food; = primary producer
- Photosynthesis: chlorophyll uses light to turn CO2 + water into organic molecules
- No sunlight? Some bacteria use hydrogen sulfide + CO2 · autotrophs are the foundation of all ecosystems
- Herbivore: plants only (cow, rabbit, grasshopper)

Flow of Energy in an Ecosystem — تدفق الطاقة في النظام البيئي

2.2 · The 10% Rule — نماذج تدفق الطاقة

- Food chain: one path of energy flow; arrows = energy (flower → grasshopper → mouse → snake)
- Food web: many interconnected food chains; most organisms eat more than one species
- At each step some energy is used for cell processes, some released to the environment
- Each step in a chain or web = a trophic level

MEMORY HOOKS

- **Auto = self** · Autotroph = self-feeder — ذاتي
- **Hetero = other** · Heterotroph = feeds on others — غيره
- **Arrow = 'is eaten by'** · Arrows point to the eater, the way energy goes — السهم للآكل
- **Divide by 10** · Each level up: move the decimal one place left — قسمة على 10

WORDS TO KNOW

Autotroph ذاتي التغذية	**Heterotroph** غير ذاتي التغذية	**Herbivore** آكل النباتات	**Carnivore** آكل اللحوم
Omnivore القارت	**Detritivore** آكل الفتات	**Decomposer** المحلل	**Food chain** السلسلة الغذائية
Food web الشبكة الغذائية	**Trophic level** المستوى الغذائي	**Ecological pyramid** الهرم البيئي	**Biomass** الكتلة الحية
Pyramid of numbers هرم الأعداد			

In the Catalyst mock exam: Q5, Q6, Q27 · scan the QR for this lesson's full test

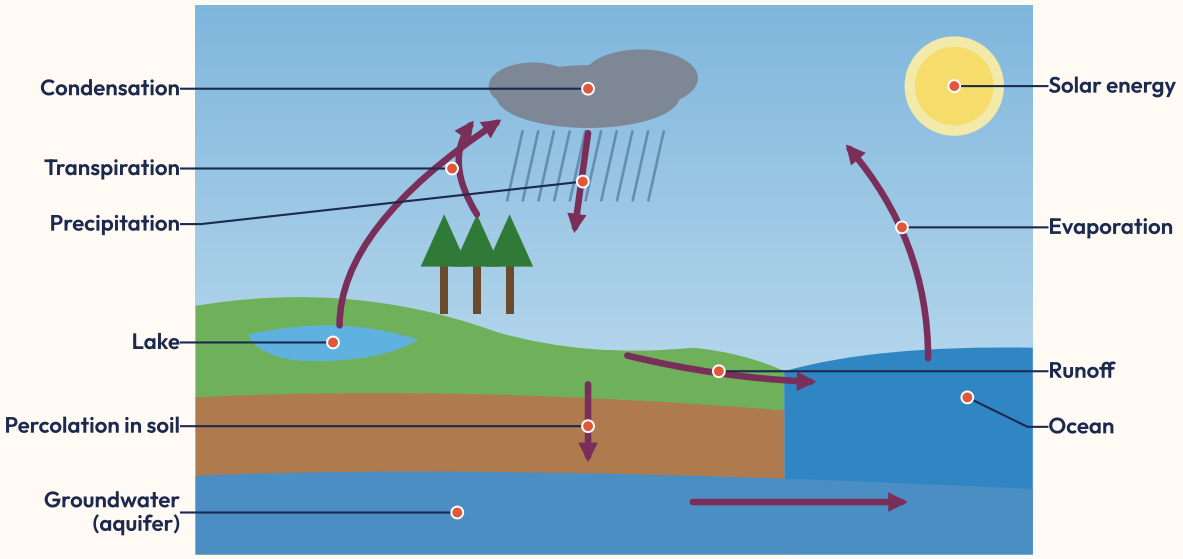
MODULE 2 · LESSON 3 · PART 3.1 · BOOK PP. 19–20

Round & Round

Cycles in the Biosphere · The Water Cycle الدورات في المحيط الحيوي · دورة الماء

SCAN FOR PRACTICE

MUST KNOW



1. **Law of conservation of mass**: matter is not created or destroyed → new life is built from **existing atoms**
   المادة لا تفنى ولا تستحدث

2. **Nutrient** = chemical substance an organism must get from its environment (C, N, P, water)
   العناصر الغذائية

3. **Biogeochemical cycle** = exchange of matter through the biosphere: bio + geo + chemical
   دورة بيوجيوكيميائية

4. **Evaporation**: the Sun's energy turns water into vapor (90% from oceans, lakes, rivers)
   تبخر 90%

5. **Transpiration**: about 10% of vapor comes from plant surfaces
   نتح 10%

6. **Condensation** → clouds · **precipitation** → rain/snow · **runoff** → rivers, oceans · **percolation** → groundwater
   تكاثف · هطول · جريان · تسرب

Process · العملية	Direction · الاتجاه	Key fact · معلومة
Evaporation · تبخر	surface → air	90% of vapor
Transpiration · نتح	plants → air	10% of vapor
Condensation · تكاثف	vapor → clouds	around dust
Precipitation · هطول	clouds → ground	rain, snow
Runoff / Percolation	over / into land	→ oceans / groundwater

DON'T FALL FOR IT

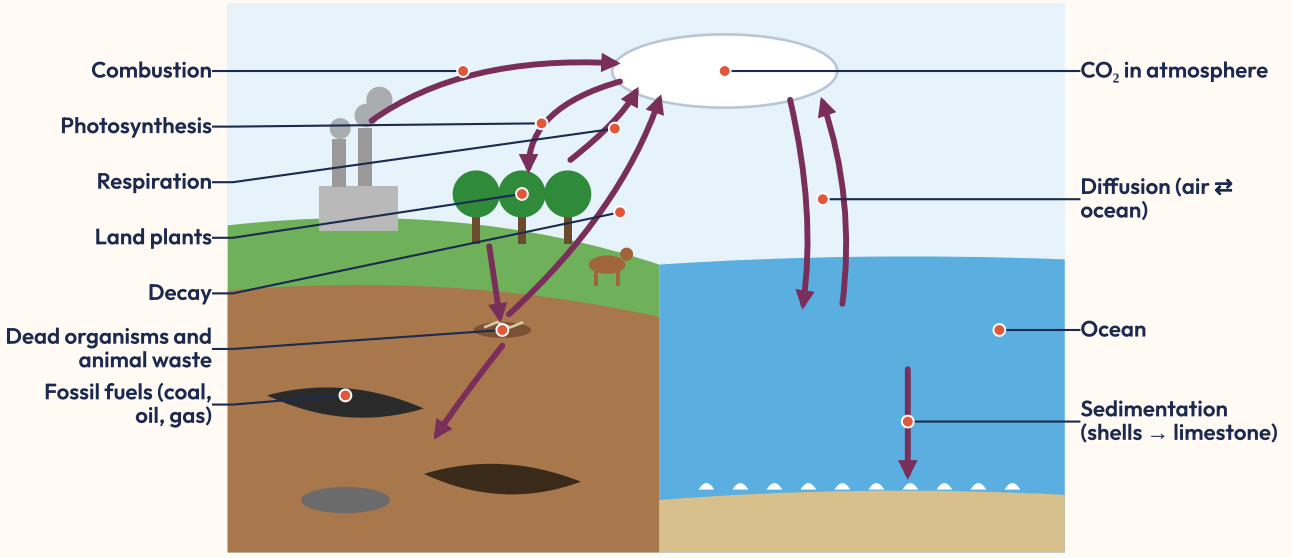
× Most water vapor comes from plants.	✓ 90% comes from oceans, lakes and rivers.	Transpiration is only about 10%. النتح 10% بس.
× Condensation = rain falling.	✓ Condensation forms clouds; falling = precipitation.	Vapor cools into droplets. التكاثف = سحب.
× Energy cycles like matter.	✓ Matter cycles; energy flows one way.	Energy is lost as heat. الطاقة لا تدور.

Carbon Loop

The Carbon and Oxygen Cycles دورتا الكربون والأكسجين

SCAN FOR PRACTICE

MUST KNOW



1. **Photosynthesis**: plants + algae turn CO_2 + water → **carbohydrates** and release **oxygen**
تشيل CO2

2. **Cellular respiration**: organisms use oxygen and release CO_2
ترجع CO2

3. Decay of dead organisms releases CO_2; decomposition puts carbon into soil
التحلل

4. Buried organic matter → **coal, oil, gas**; **combustion** returns CO_2
وقود أحفوري

5. C + O + **calcium** → **calcium carbonate** (shells → limestone, white cliffs of Dover)
كربونات الكالسيوم

6. Released again only by **weathering and erosion**
التجوية والتعرية

Process · العملية	CO_2 in air · في الجو	Cycle · الدورة
Photosynthesis · بناء ضوئي	removes ↓	short-term
Respiration · تنفس	adds ↑	short-term
Decay · تحلل	adds ↑	short-term
Combustion · احتراق	adds ↑	long-term (fossil fuels)
Sedimentation · ترسيب	stores	long-term (limestone)

DON'T FALL FOR IT

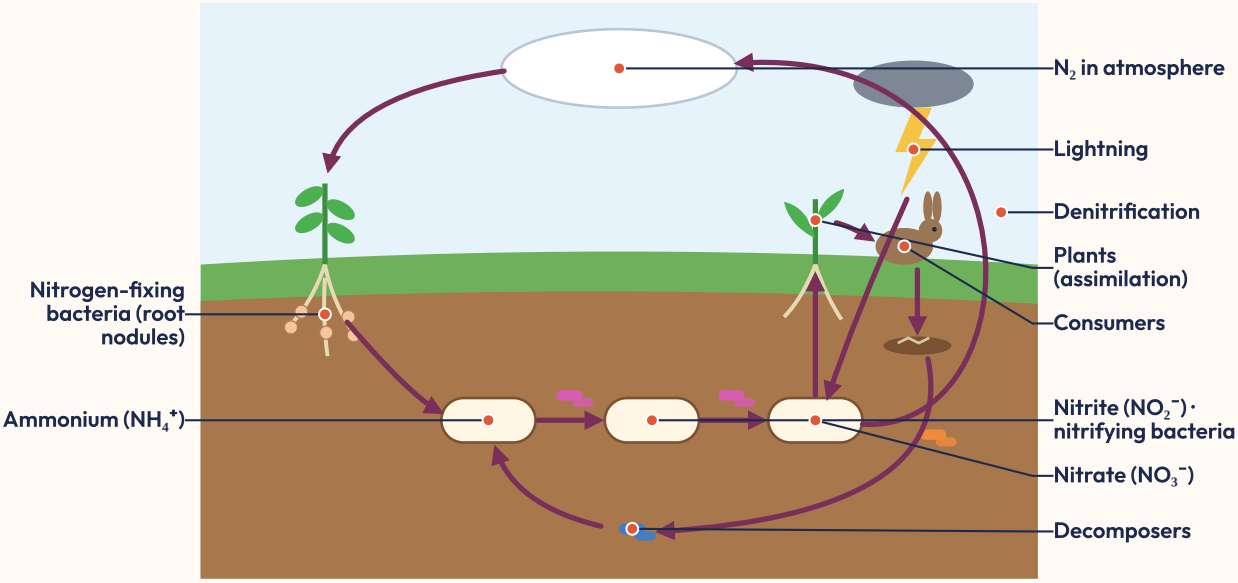
✕ Plants only do photosynthesis.	✓ Plants also respire and release CO_2.	Plants respire all the time. النبات بيتنفس كمان.
✕ Combustion is part of the short-term cycle.	✓ It releases long-term (fossil) carbon.	Fossil fuels took millions of years. طويل المدى.
✕ Limestone carbon is lost forever.	✓ Weathering and erosion release it.	It stays until rocks wear away. يرجع بالتجوية.

Fix the Air

The Nitrogen Cycle دورة النيتروجين

SCAN FOR PRACTICE

MUST KNOW



1. Most nitrogen is in the **atmosphere** as N_2, which plants and animals can't use directly
   أغلبه في الجو
2. **Fixation**: N_2 → **ammonium NH_4^+**, by specialized **bacteria** (root nodules, soil)
   بكتيريا مثبتة
3. **Lightning** can also make nitrogen-oxygen compounds
   البرق
4. **Nitrification**: NH_4^+ → **NO_2^-** and **NO_3^-**
   النترجة
5. Plants absorb NO_3^- through their roots (**assimilation**) to make **proteins**; nitrogen moves through the food web
   الامتصاص (التمثيل) ← بروتينات
6. Back to soil: animal wastes + decomposition of dead matter into **ammonia**
   يرجع للتربة
7. **Denitrification**: soil bacteria turn NO_2^-/NO_3^- → N_2 back to the air
   يرجع للجو
8. Fertilizer runoff → **algae blooms**
   تكاثر الطحالب
9. Factory nitrogen-oxygen compounds + water → **acid rain**
   مطر حمضي

Process · العملية	Change · التحويل	Done by · بواسطة
Fixation · تثبيت	N_2 → NH_4^+	bacteria; lightning
Nitrification · نترجة	NH_4^+ → NO_2^- → NO_3^-	nitrifying bacteria
Assimilation · تمثيل	NO_3^- → plant proteins	plants
Denitrification · عكس النترجة	NO_2^-/NO_3^- → N_2	denitrifying bacteria

DON'T FALL FOR IT

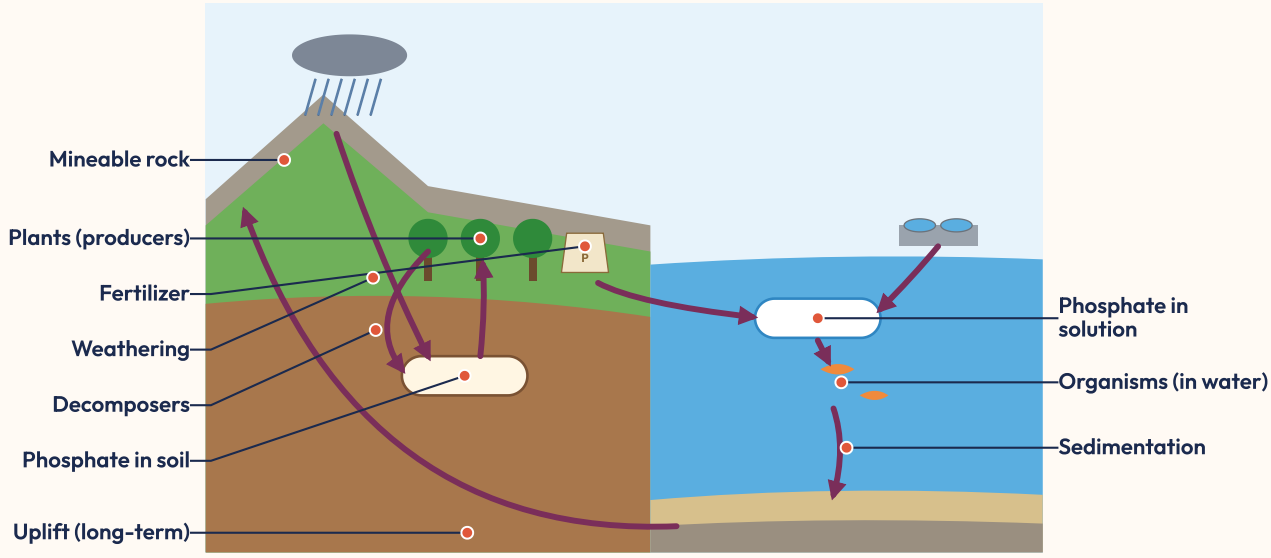
× Plants absorb N_2 from the air.	✓ Plants absorb fixed nitrogen (NO_3^-) from soil.	N_2 must be fixed first. النبات ياخد النترات.
× Nitrification = fixation.	✓ Fixation N_2 → NH_4^+; nitrification NH_4^+ → NO_3^-.	Two different steps. خطوتين مختلفتين.
× Denitrification adds nitrogen to soil.	✓ It removes it, returning N_2 to the air.	De = back out. يرجعه للجو.

Rock to Root

The Phosphorus Cycle دورة الفوسفور

SCAN FOR PRACTICE

MUST KNOW



1. Phosphorus is essential for **growth and development**
النمو والتطور

2. Phosphate in soil → **producers** → **consumers**
تربة ← منتج ← مستهلك

3. Wastes and dead organisms → **decomposers** return phosphate to the soil
المحللات

4. **Precipitation and sedimentation** → phosphate forms **rocks**
ترسيب ← صخور

5. Geological changes and **uplift** move sediment to land
الرفع

6. **Weathering / erosion** slowly add phosphate back to soil
التجوية

7. Phosphate **mining**, **sewage** treatment, **fertilizer** production add phosphate
تعدين · صرف · سماد

8. Too much → **algae blooms** that upset the balance (fresh or marine water)
تكاثر الطحالب

9. Carbon AND phosphorus both have short- and long-term cycles
الكربون والفوسفور

	Short-term · قصيرة	Long-term · طويلة
Where · أين	soil, producers, consumers	sediment, rock
Moved by · بواسطة	eating, decomposers	sedimentation, uplift, weathering
Speed · السرعة	fast	very slow

DON'T FALL FOR IT

✕ Phosphorus cycles through the atmosphere.	✓ It cycles through rock, soil, water and organisms.	No gas phase in the figure. مفيش غاز في الجو.
✕ Sediment phosphate is quickly reused on land.	✓ Only after uplift and weathering, slowly.	That is the long-term cycle. بيطء.
✕ More phosphate in a pond = healthier pond.	✓ It causes algae blooms.	Blooms upset the balance. طحالب زيادة.

Cycling of Matter

Book pp. 19–24 دورة المادة

LESSON TEST

MIND MAP · THE WHOLE LESSON

3.1 · Round & Round · الدورات في المحيط الحيوي

- Law of conservation of mass: matter is not created or destroyed → new life is built from existing atoms
- Nutrient = chemical substance an organism must get from its environment (C, N, P, water)
- Biogeochemical cycle = exchange of matter through the biosphere: bio + geo + chemical

3.2 · Carbon Loop · دورتا الكربون والأكسجين

- Photosynthesis: plants + algae turn CO2 + water → carbohydrates and release oxygen
- Cellular respiration: organisms use oxygen and release CO2
- Decay of dead organisms releases CO2; decomposition puts carbon into soil

Cycling of Matter دورة المادة

3.3 · Fix the Air · دورة النيتروجين

- Most nitrogen is in the atmosphere as N2, which plants and animals can't use directly
- Fixation: N2 → ammonium NH4+, by specialized bacteria (root nodules, soil)
- Lightning can also make nitrogen-oxygen compounds

3.4 · Rock to Root · دورة الفوسفور

- Phosphorus is essential for growth and development
- Phosphate in soil → producers → consumers
- Wastes and dead organisms → decomposers return phosphate to the soil

MEMORY HOOKS

E · C · P · Evaporate, Condense, Precipitate: the core loop
تبخر · تكاثف · هطول

Photo takes, Resp gives · Photosynthesis takes CO_2, respiration gives it back
ياخد ويرجع

Fix · Nitri · Deni · In → change → out: fixation, nitrification, denitrification
دخول ← تغيير ← خروج

Rock → Root · Phosphorus comes from rocks, not the air
من الصخر

Trans = through the plant · Transpiration leaves through plant surfaces
النتح من النبات

Fossil = old carbon · Coal, oil, gas = long-term carbon store
مخزن طويل

De = back · DEnitrification sends N_2 back up
رجوع

Slow rock, fast life · Long-term = rocks (slow) · short-term = living things (fast)
الصخر بطيء

WORDS TO KNOW

Nutrient عنصر غذائي	**Biogeochemical cycle** الدورة البيوجيوكيميائية	**Matter** المادة	**Evaporation** التبخر
Transpiration النتح	**Condensation** التكاثف	**Precipitation** الهطول	**Percolation** التسرب
Photosynthesis البناء الضوئي	**Cellular respiration** التنفس الخلوي	**Fossil fuels** الوقود الأحفوري	**Combustion** الاحتراق
Calcium carbonate كربونات الكالسيوم	**Decay** التحلل	**Nitrogen fixation** تثبيت النيتروجين	**Nitrification** النترجة
Assimilation التمثيل (الامتصاص)	**Denitrification** عكس النترجة	**Algae bloom** تكاثر الطحالب	**Root nodules** العقد الجذرية
Acid rain المطر الحمضي	**Phosphorus** الفوسفور	**Sedimentation** الترسيب	**Uplift** الرفع الجيولوجي
Weathering التجوية	**Phosphate mining** تعدين الفوسفات		

In the Catalyst mock exam: Q7, Q8, Q9, Q10, Q26 · scan the QR for this lesson's full test

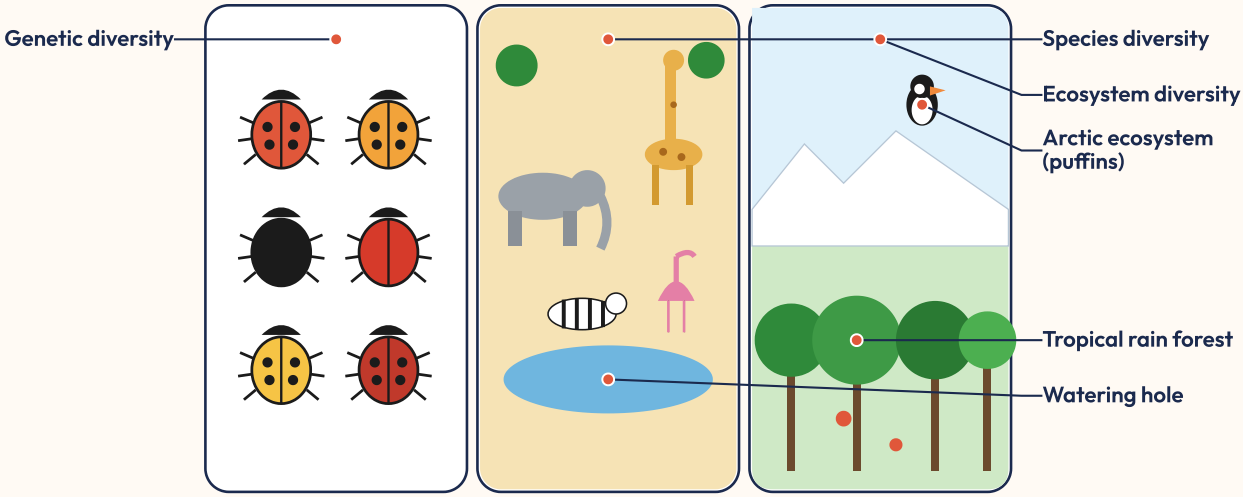
MODULE 5 · LESSON 1 · PART 1.1 · BOOK PP. 30–32

Variety of Life

What Is Biodiversity? ما هو التنوع الحيوي؟

SCAN FOR PRACTICE

MUST KNOW



1. **Biodiversity** = variety of life in an area, set by the number of different species
تنوع الحياة

2. Increased by **new species** · decreased by **extinction** (last member dies)
يزيد ويقل

3. Increases ecosystem **stability** and the health of the biosphere
الاستقرار

4. **Genetic**: variety of genes in a population (ladybird colors, disease resistance) → better survival
جيني

5. **Species**: number and relative abundance of species (watering hole); increases toward the **equator**
أنواع

6. **Ecosystem**: variety of ecosystems (Arctic puffins vs South American rainforest)
أنظمة

Type · النوع	Means · المعنى	Example · مثال
Genetic · جيني	variety of genes in one population	ladybird beetle colors
Species · أنواع	number + abundance of species	watering hole
Ecosystem · أنظمة	variety of ecosystems	Arctic vs rainforest

DON'T FALL FOR IT

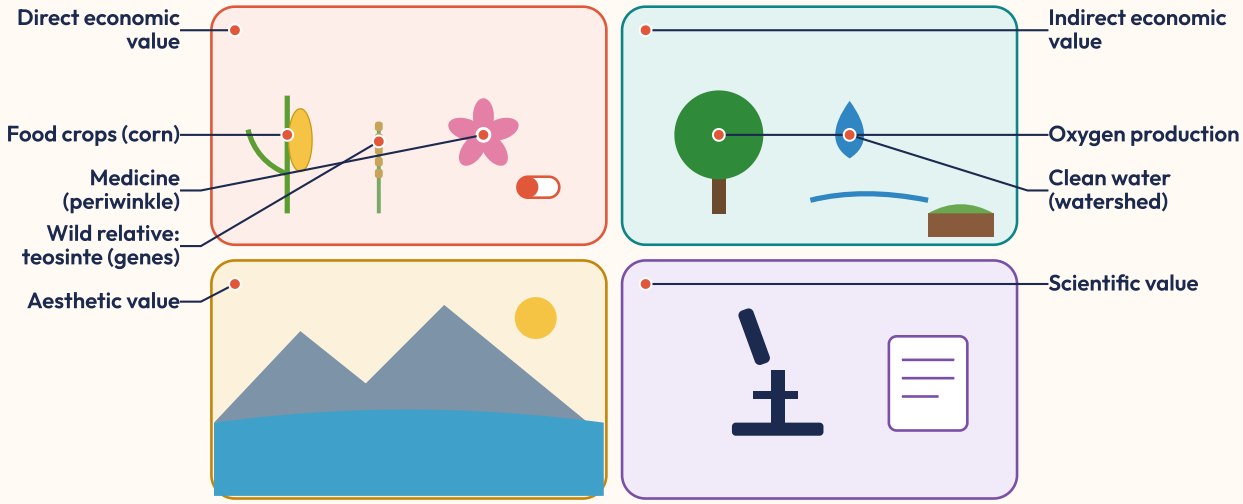
✕ Different-colored beetles of one species = species diversity.	✓ That is genetic diversity.	Same species → variety of genes. نفس النوع = جيني.
✕ Biodiversity is the same everywhere.	✓ It increases toward the equator.	Tropics have the most species. يزيد عند الاستواء.
✕ Biodiversity = number of animals.	✓ It includes all life and genetic variety.	Plants and microbes count too. كل الكائنات.

Worth Saving

The Importance of Biodiversity أهمية التنوع الحيوي

SCAN FOR PRACTICE

MUST KNOW



1. Food, clothing, energy, **medicine**, shelter
منتجات

2. Wild relatives = reservoirs of genes: **teosinte** → virus-resistant corn
التيوسنت

3. Medicines: **penicillin** (bread mold) · aspirin (willow) · **Madagascar periwinkle** → leukemia (20% → 95%+ survival)
أدوية

4. Plants give O_2 and remove CO_2; safe drinking water; nutrient cycling
خدمات

5. Protect against floods and drought, make fertile soil, decompose wastes, regulate climate
حماية

6. New York watersheds: ~$1.5 billion vs >$6 billion filter → nature is **cheaper** than technology
الطبيعة أرخص

7. **Aesthetic**: landscapes for recreation/inspiration (Lake Tahoe's Emerald Bay)
جمالي

8. **Scientific**: study of the environment brings groups together, new technologies
علمي

Value · القيمة	Meaning · المعنى	Example · مثال
Direct economic · مباشرة	products we use	food, medicine, wood
Indirect economic · غير مباشرة	ecosystem services	O_2, clean water, soil
Aesthetic · جمالية	beauty, recreation	Lake Tahoe
Scientific · علمية	knowledge, technology	studying ecosystems

DON'T FALL FOR IT

× Oxygen from forests is a direct value.	✓ It is indirect (a service).	Direct = products we use. الأكسجين خدمة.
× Species with no use now can be lost.	✓ Their genes may be needed later.	Future value can rise. ممكن نحتاجها بعدين.
× Technology is always cheaper than nature.	✓ NYC: nature $1.5 bn vs filter $6 bn.	Healthy ecosystems are cheaper. الطبيعة أرخص.

Biodiversity

Book pp. 30–36 التنوع الحيوي

LESSON TEST

MIND MAP · THE WHOLE LESSON

1.1 · Variety of Life — ما هو التنوع الحيوي؟

- Biodiversity = variety of life in an area, set by the number of different species
- Increased by new species · decreased by extinction (last member dies)
- Increases ecosystem stability and the health of the biosphere
- Genetic: variety of genes in a population (ladybird colors, disease resistance) → better survival

Biodiversity التنوع الحيوي

1.2 · Worth Saving — أهمية التنوع الحيوي

- Food, clothing, energy, medicine, shelter
- Wild relatives = reservoirs of genes: teosinte → virus-resistant corn
- Medicines: penicillin (bread mold) · aspirin (willow) · Madagascar periwinkle → leukemia (20% → 95%+ survival)
- Plants give O2 and remove CO2; safe drinking water; nutrient cycling

MEMORY HOOKS

G·S·E · Genes → Species → Ecosystems: small to big
من الصغير للكبير

Closer to the equator, more species · Biodiversity rises toward the tropics
عند الاستواء أكثر

Direct = you can BUY it · Food, wood, medicine
تقدري تشتريها

Indirect = nature does a JOB · Oxygen, clean water, soil
الطبيعة بتشتغل

WORDS TO KNOW

Biodiversity التنوع الحيوي	**Extinction** الانقراض	**Genetic diversity** التنوع الجيني	**Species diversity** تنوع الأنواع
Ecosystem diversity تنوع الأنظمة البيئية	**Direct economic value** القيمة الاقتصادية المباشرة	**Indirect economic value** القيمة الاقتصادية غير المباشرة	**Watershed** مستجمع المياه
Aesthetic value القيمة الجمالية	**Genetic engineering** الهندسة الوراثية	**Teosinte** التيوسنت	

In the Catalyst mock exam: Q11, Q12 · scan the QR for this lesson's full test

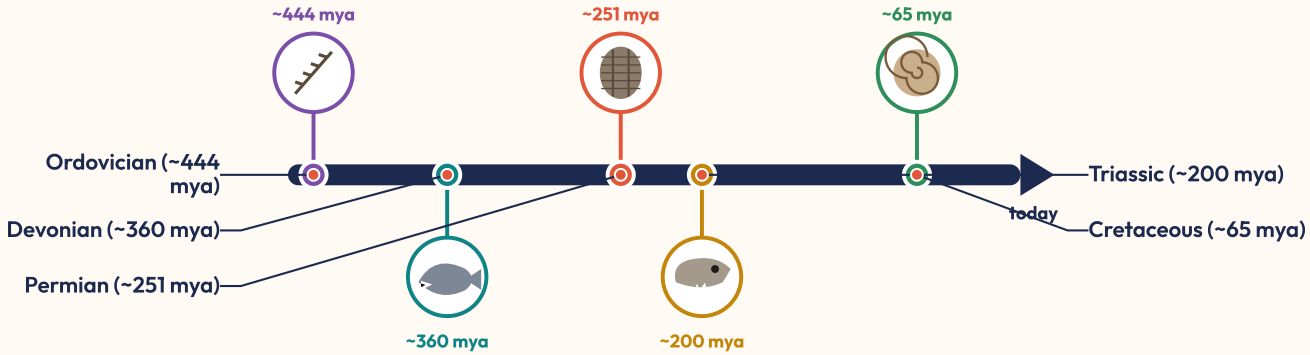
MODULE 5 · LESSON 2 · PART 2.1 · BOOK PP. 37–38

Gone Forever

Extinction Rates معدلات الانقراض

SCAN FOR PRACTICE

MUST KNOW



1. Species go extinct when they can't survive and reproduce after **too fast or drastic** change
تغيير سريع

2. **Background extinction**: the gradual, natural process
تدريجي طبيعي

3. **Mass extinction**: a large % of all species lost in a short time; last one **~65 mya**
جماعي

4. Ordovician ~444 · Devonian ~360 · Permian ~251 · Triassic ~200 · Cretaceous ~65 million years ago
الترتيب

5. Today: **~1000×** the background rate (some scientists) → many think we are in a mass extinction
1000 ضعف

6. Most predicted extinctions near the **equator**
عند الاستواء

7. 60% of mammal and **81%** of bird extinctions (past 500 years) were on islands
81% طيور

8. No natural predators during evolution → can't escape introduced dogs, cats, rats, humans
مفيش مفترسات

9. Small populations + rarely travel between islands + no resistance to new diseases
جماعات صغيرة

Period · الفترة	Time · الزمن	Example · مثال
Ordovician	~444 mya	graptolites
Devonian	~360 mya	armored fish
Permian	~251 mya	trilobite
Triassic	~200 mya	Cynognathus
Cretaceous	~65 mya	ammonite

DON'T FALL FOR IT

× Scientists worry about background extinction.	✓ They worry about the recent INCREASE in rate.	Background is natural. القلق من الزيادة.
× The last mass extinction was 444 mya.	✓ It was about 65 mya (Cretaceous).	444 mya was the oldest of the five. 65 مليون.
× Island species are safer because they're isolated.	✓ They are MORE vulnerable.	No predators, small populations, no resistance. أكثر عرضة.

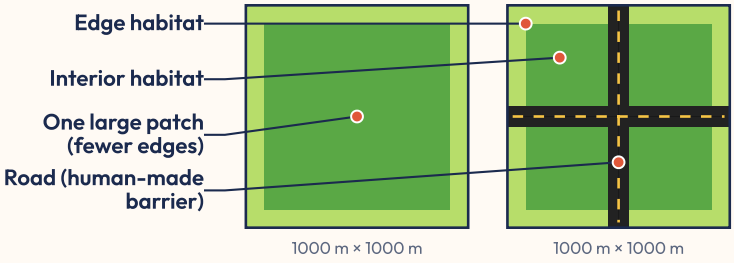
MODULE 5 · LESSON 2 · PART 2.2 · BOOK PP. 39–42

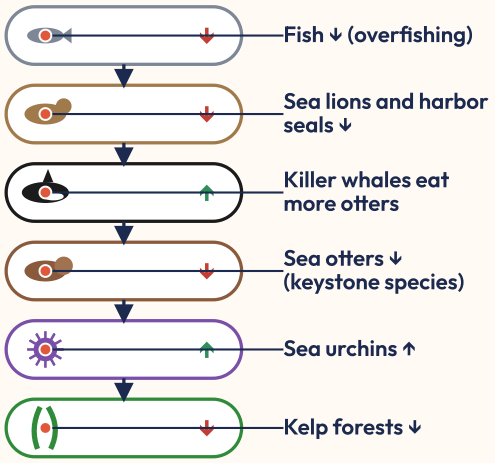
Lost Ground

Overexploitation and Habitat Loss الاستغلال المفرط وفقدان الموطن

SCAN FOR PRACTICE

MUST KNOW





1. Habitat loss · pollution · invasive species · overexploitation · climate change
خمس تهديدات

2. Humans change conditions **faster than new traits can evolve**
أسرع من التطور

3. **Natural resources** = all materials and organisms in the biosphere (minerals, fuels, plants, soil, water, air, solar energy)
الموارد الطبيعية

4. Excessive use of species with economic value
استخدام زائد

5. Mahogany (~100 years to mature) · bison 50 million → <1000 (1889) · passenger pigeons extinct (early 1900s)
أمثلة

6. Ocelot (fur, habitat loss) · rhinos (horns); historically the #1 cause
كان السبب الأول

7. **Destruction**: clearing tropical rain forests (more than half of all species live there)
التدمير

8. **Disruption**: Alaska chain → **keystone species** (sea otter) · fish ↓ → seals ↓ → orcas eat otters → urchins ↑ → kelp ↓
النوع المفتاحي

9. **Fragmentation**: small parcels → fewer species, less genetic diversity; more **edge effects** (hotter, windier, drier edges; predators/parasites)
التجزئة وتأثير الحافة

Threat · التهديد	Meaning · المعنى	Example · مثال
Overexploitation · استغلال مفرط	excessive use	bison, passenger pigeon
Destruction · تدمير	habitat removed	rain forest cleared
Disruption · اضطراب	food web changed	otters → kelp
Fragmentation · تجزئة	habitat split	roads → edges

DON'T FALL FOR IT

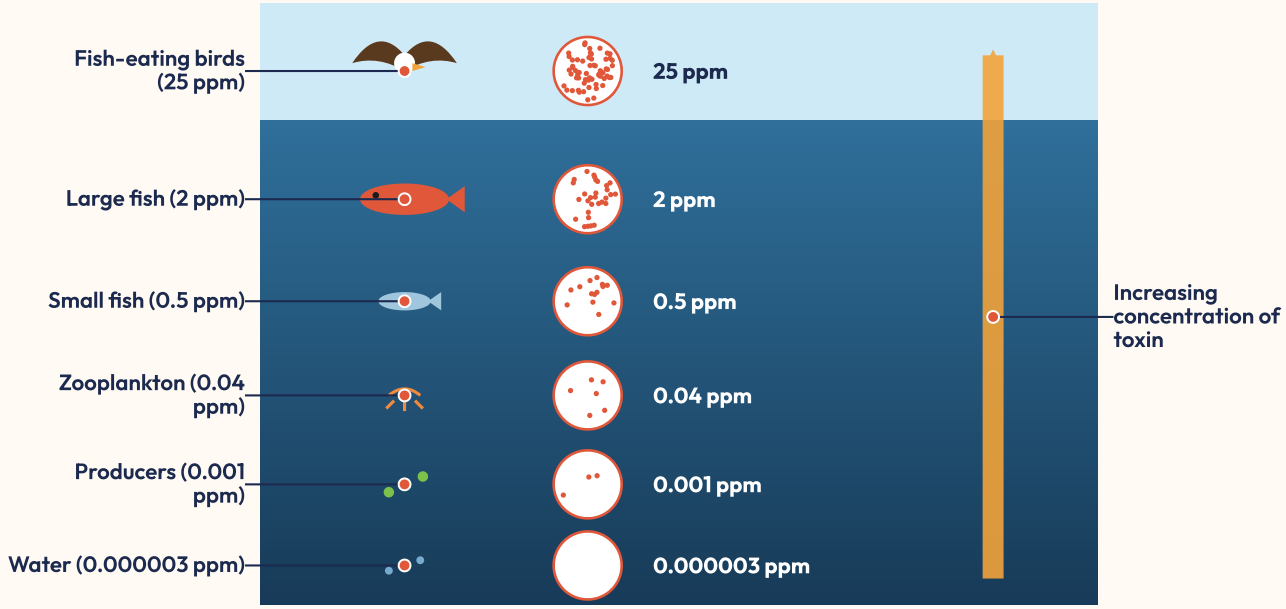
× Overexploitation is the #1 cause today.	✓ Habitat destruction is #1 today.	Overexploitation was #1 historically. اليوم: تدمير الموطن.
× Edges are cooler and more humid.	✓ Edges are hotter, windier and drier.	In a tropical forest. الحافة أسخن وأجف.
× Fewer otters → more kelp.	✓ Fewer otters → more urchins → less kelp.	Urchins eat kelp. القنافذ تأكل الكيلب.

Poison Climb

Climate Change, Pollution and Invasive Species تغير المناخ والتلوث والأنواع الغازية

SCAN FOR PRACTICE

MUST KNOW



1. Species evolved for certain temperature ranges; those that can't adapt **die**
   نطاق حرارة
2. Expected to overtake habitat destruction as the **#1 threat** (first half of this century)
   هيبقى التهديد الأول
3. May threaten ~**25%** of land species by 2050
   25%
4. DDT, PCBs accumulate in tissues → concentration **increases** up each trophic level
   يزيد مع المستوى
5. Water 0.000003 · producers 0.001 · zooplankton 0.04 · small fish 0.5 · large fish 2 · birds **25 ppm**
   الأرقام
6. DDT → thin eggshells; bald eagle and peregrine falcon nearly extinct; DDT banned in some places
   قشر بيض رقيق
7. Fossil fuels → SO_2, cars → nitrogen oxides → sulfuric/nitric acid; removes Ca, K from soil
   مطر حمضي
8. **Eutrophication**: N + P from fertilizer, waste, sewage → algae boom → O_2 used up → suffocation
   الإثراء الغذائي
9. Natural, but humans speed it up
   طبيعي بس الإنسان بيسرعه

Threat · التهديد	Cause · السبب	Effect · الأثر
Biomagnification · تضخم حيوي	DDT, PCBs	top predators most affected
Acid precipitation · هطول حمضي	SO_2, nitrogen oxides	soil loses Ca, K; fish die
Eutrophication · إثراء غذائي	N + P runoff	algae use up O_2
Invasive species · غازية	no natural controls	~40% of extinctions since 1750

DON'T FALL FOR IT

× Producers have the most toxin.	✓ Top predators have the most (25 ppm).	Concentration rises each level. الأعلى أكثر.
× Eutrophication adds oxygen because algae photosynthesize.	✓ It USES UP oxygen.	Algae growth and decay remove O_2. يستهلك الأكسجين.
× Every introduced species is invasive.	✓ Only those that disrupt the habitat.	Not all introduced species become invasive. مش كلهم.

Threats to Biodiversity

Book pp. 37–44 التهديدات للتنوع الحيوي

LESSON TEST

MIND MAP · THE WHOLE LESSON

2.1 · Gone Forever — معدلات الانقراض

- Species go extinct when they can't survive and reproduce after too fast or drastic change
- Background extinction: the gradual, natural process
- Mass extinction: a large % of all species lost in a short time; last one ~65 mya
- Ordovician ~444 · Devonian ~360 · Permian ~251 · Triassic ~200 · Cretaceous ~65 million years ago

2.2 · Lost Ground — الاستغلال المفرط وفقدان ال

- Habitat loss · pollution · invasive species · overexploitation · climate change
- Humans change conditions faster than new traits can evolve
- Natural resources = all materials and organisms in the biosphere (minerals, fuels, plants, soil, water, air, solar energy)
- Excessive use of species with economic value

Threats to Biodiversity التهديدات للتنوع الحيوي

2.3 · Poison Climb — تغير المناخ والتلوث والأنو

- Species evolved for certain temperature ranges; those that can't adapt die
- Expected to overtake habitat destruction as the #1 threat (first half of this century)
- May threaten ~25% of land species by 2050
- DDT, PCBs accumulate in tissues → concentration increases up each trophic level

MEMORY HOOKS

★ **O·D·P·T·C** · Old Dogs Play Tag Cheerfully: Ordovician, Devonian, Permian, Triassic, Cretaceous
ترتيب الانقراضات

★ **444 → 65** · First and last of the five mass extinctions
أول وآخر

★ **H·I·P·P·O** · Habitat loss, Invasive species, Pollution, Population/climate, Overexploitation
التهديدات

★ **Then: over-use · Now: habitat** · Overexploitation was #1, habitat destruction is #1 today
زمان واليوم

★ **Higher = more poison** · Toxins pile up at the top of the food chain
كل ما تطلعي السم يزيد

★ **Eu-troph = well fed** · Too much food (N, P) for algae
أكل زيادة للطحالب

WORDS TO KNOW

Background extinction الانقراض الخلفي	**Mass extinction** الانقراض الجماعي	**Extinction** الانقراض	**Paleontologist** عالم الحفريات
Island species أنواع الجزر	**Anthropogenic change** تغيير بشري المنشأ	**Natural resources** الموارد الطبيعية	**Overexploitation** الاستغلال المفرط
Keystone species النوع المفتاحي	**Habitat fragmentation** تجزئة الموطن	**Edge effects** تأثيرات الحافة	**Biological magnification** التضخم الحيوي
Eutrophication الإثراء الغذائي	**Acid precipitation** الهطول الحمضي	**Invasive species** الأنواع الغازية	**Introduced species** الأنواع المُدخلة
DDT مبيد DDT			

In the Catalyst mock exam: Q13, Q14, Q15, Q29 · scan the QR for this lesson's full test

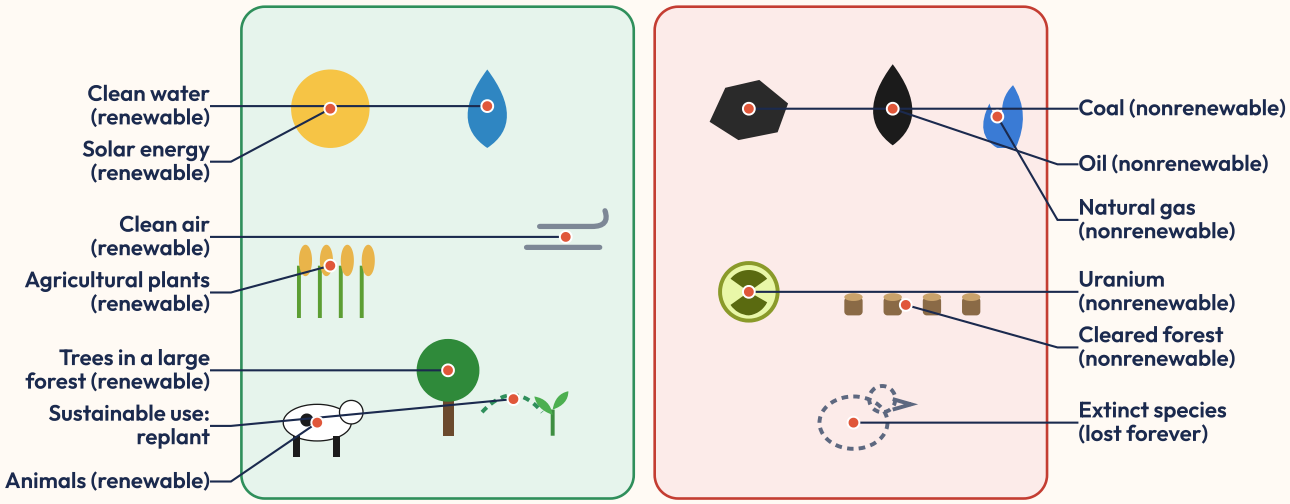
MODULE 5 · LESSON 3 · PART 3.1 · BOOK PP. 45–47

Use It Wisely

Natural Resources الموارد الطبيعية

SCAN FOR PRACTICE

MUST KNOW



1. Growing population + higher consumption → need a **long-term plan**
   خطة طويلة
2. Consumption per person is much higher in **developed** countries
   الدول المتقدمة
3. More industry → higher standard of living → more consumption
   صناعة أكثر = استهلاك أكثر
4. **Renewable**: replaced faster than consumed (solar, crops, animals, clean water, clean air), but can be **depleted**
   متجددة
5. **Nonrenewable**: limited or replaced over extremely long periods (coal, gas, oil, uranium)
   غير متجددة
6. Species are renewable until **extinct**; a few trees = renewable, a whole cleared forest = not (depends on **context**)
   حسب السياق
7. Use resources at a rate they can be **replaced or recycled** while keeping the biosphere healthy
   بمعدل التجدد
8. Conservation = **reduce**, **recycle**, **preserve** ecosystems, use responsibly
   قلل · أعد التدوير · احم
9. Done by individuals, companies or countries; weigh cost, safety, reliability, aesthetics
   أفراد وشركات ودول

	Renewable · متجدد	Nonrenewable · غير متجدد
Replaced · التعويض	faster than used	very slowly / limited
Examples · أمثلة	solar, water, air, crops	coal, oil, gas, uranium
Can run out? · ينفد؟	yes, if overused	yes

DON'T FALL FOR IT

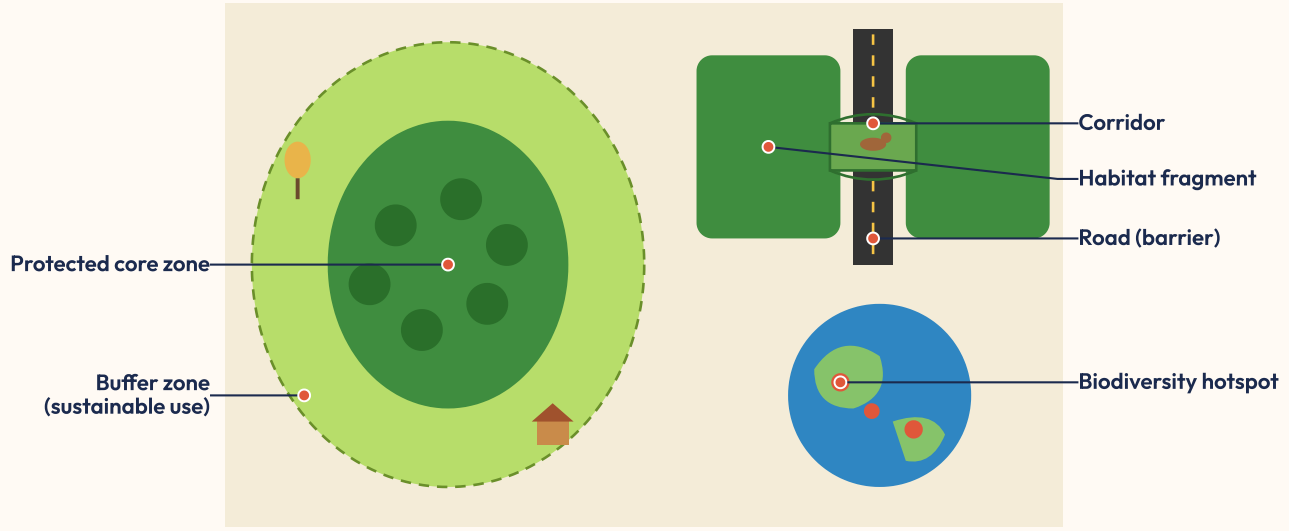
× Renewable means it can never run out.	✓ It can be depleted if demand exceeds supply.	Replacement must keep up with use. ممكن ينفد.
× Trees are always renewable.	✓ A whole cleared forest is nonrenewable.	It depends on the context. حسب السياق.
× Developing countries consume the most per person.	✓ Developed countries do.	More industry, higher consumption. المتقدمة أكثر.

Safe Zones

Protecting Biodiversity حماية التنوع الحيوي

SCAN FOR PRACTICE

MUST KNOW



1. USA: **Yellowstone** (1872), first national park
يلوستون

2. **~10–15%** of world's land is reserve; small reserves suffer from human activity
15%–10

3. Costa Rica **megareserves**: protected core + **buffer zones** (sustainable use allowed)
المحميات الكبرى

4. **Endemic species** = found only in one area
متوطن

5. Hotspot: ≥ **1500** endemic vascular plants AND lost ≥ **70%** of original habitat; **36** hotspots
الشروط

6. **~half** of all species; covered 17% of surface, only ~a tenth remains; debate: wetlands give services too
نصف الأنواع

7. **Corridors** connect fragments → more species and genetic variation; but diseases spread + more edges
الممرات

8. **Endangered Species Act** (USA, 1973)
قانون الأنواع المهددة

9. **CITES** (1973): bans trade of endangered species and parts (ivory, rhino horns)
سايتس

Method · الطريقة	How it helps · الفائدة	Problem · المشكلة
Small reserve · محمية صغيرة	protects habitat	heavy human impact
Megareserve · محمية كبرى	protects + allows sustainable use	needs large land
Corridor · ممر	connects fragments	spreads disease, more edges
Hotspots · نقاط ساخنة	saves the most species	ignores other areas

DON'T FALL FOR IT

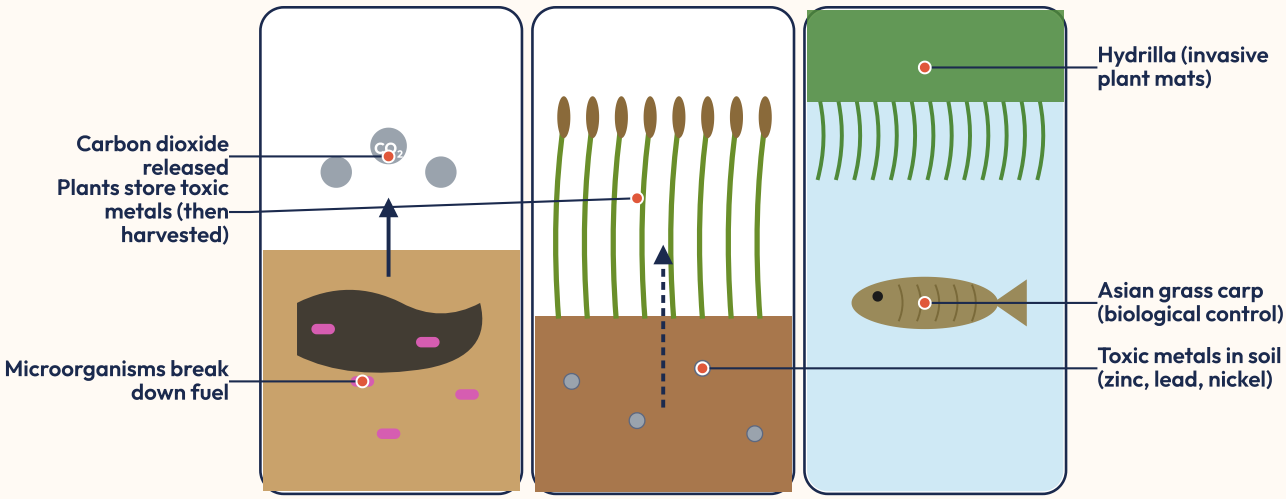
× Corridors solve habitat destruction.	✓ They help but spread disease and add edges.	One large habitat has fewer edges. مش حل كامل.
× Hotspot = any place with many species.	✓ ≥1500 endemic plants AND ≥70% habitat lost.	Both rules are needed. الشرطين مغا.
× Buffer zones allow unlimited use.	✓ Only SUSTAINABLE use.	The core stays protected. استخدام مستدام فقط.

Heal the Land

Restoring Ecosystems استعادة الأنظمة البيئية

SCAN FOR PRACTICE

MUST KNOW



1. Mining, oil spills, toxic spills, cleared rain forest soil can leave land unable to support life
   أسباب الضرر
2. Recovery time depends on the **size of the area** and the **type of disturbance**
   المساحة ونوع الضرر
3. Larger area → longer recovery; not directly linked to natural vs human-made
   أكبر = أطول
4. Using organisms (prokaryotes, fungi, plants) to **detoxify** a polluted area
   إزالة السموم
5. 1975 South Carolina jet fuel leak: soil microbes broke fuel → **CO_2**; adding **nutrients** sped it up
   ميكروبات
6. Plants store zinc, lead, nickel → **harvested**; reed beds (bacteria + fungi); fish bones → lead harmless
   النباتات تخزن المعادن
7. Introduce a predator or competitor of the invasive species
   مفترس أو منافس
8. **Hydrilla** mats block light in Florida → **Asian grass carp** (early 1970s) eat it
   سمك الكارب
9. Too many carp destroy native plants → numbers are **monitored**
   مراقبة

Method · الطريقة	Uses · يستخدم	Example · مثال
Bioremediation (microbes) · ميكروبات	prokaryotes, fungi	jet fuel → CO_2
Bioremediation (plants) · نباتات	plants that store metals	zinc, lead, nickel
Biological control · مكافحة حيوية	predator / competitor	grass carp eat Hydrilla

DON'T FALL FOR IT

✕ Human-made disasters always take longer to recover.	✓ Recovery depends on area size and disturbance type.	Not directly on natural vs human-made. مش على مين السبب.
✕ Bioremediation = biological control.	✓ Bioremediation cleans pollution; biocontrol manages invaders.	Different goals. هدف مختلف.
✕ More carp is always better.	✓ Too many carp destroy native plants.	Numbers are monitored. لازم مراقبة.

Conserving Biodiversity

Book pp. 45–52 الحفاظ على التنوع الحيوي

LESSON TEST

MIND MAP · THE WHOLE LESSON

3.1 · Use It Wisely — الموارد الطبيعية

- Growing population + higher consumption → need a long-term plan
- Consumption per person is much higher in developed countries
- More industry → higher standard of living → more consumption
- Renewable: replaced faster than consumed (solar, crops, animals, clean water, clean air), but can be depleted

3.2 · Safe Zones — حماية التنوع الحيوي

- USA: Yellowstone (1872), first national park
- ~10–15% of world's land is reserve; small reserves suffer from human activity
- Costa Rica megareserves: protected core + buffer zones (sustainable use allowed)
- Endemic species = found only in one area

Conserving Biodiversity — الحفاظ على التنوع الحيوي

3.3 · Heal the Land — استعادة الأنظمة البيئية

- Mining, oil spills, toxic spills, cleared rain forest soil can leave land unable to support life
- Recovery time depends on the size of the area and the type of disturbance
- Larger area → longer recovery; not directly linked to natural vs human-made
- Using organisms (prokaryotes, fungi, plants) to detoxify a polluted area

MEMORY HOOKS

- **Re-new = comes back** · Renewable resources come back faster than we use them — يرجع
- **Fossil = forever gone** · Coal, oil, gas take millions of years — الوقود لا يعود
- **1500 · 70 · 36** · Hotspot rules and count — أرقام النقاط الساخنة
- **Core + buffer** · Megareserve = protect the middle, use the edge wisely — قلب ومنطقة عازلة
- **Bio-remediate = life repairs** · Organisms clean the pollution — الكائنات بتنظف
- **Bigger damage, longer wait** · Recovery time grows with the area — أكبر = أطول

WORDS TO KNOW

Renewable resource مورد متجدد	**Nonrenewable resource** مورد غير متجدد	**Sustainable use** الاستخدام المستدام	**Conservation** الحفاظ
Standard of living مستوى المعيشة	**Endemic species** الأنواع المتوطنة	**Biodiversity hotspot** النقطة الساخنة للتنوع	**Corridor** الممر البيئي
Buffer zone المنطقة العازلة	**Megareserve** المحمية الكبرى	**CITES** اتفاقية سايتس	**Bioremediation** المعالجة الحيوية
Biological control المكافحة الحيوية	**Hydrilla** الهيدريلا	**Recovery time** زمن التعافي	

In the Catalyst mock exam: Q16, Q17, Q18 · scan the QR for this lesson's full test

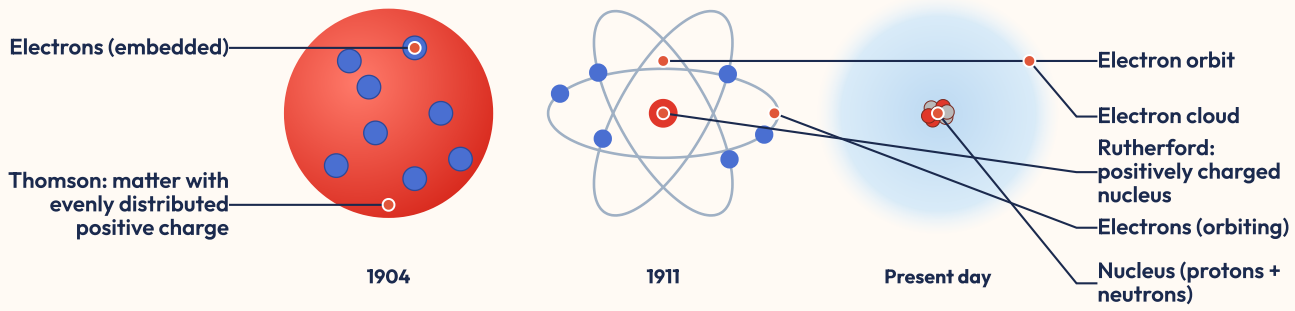
MODULE 1 · LESSON 1 · PART 1.1 · BOOK PP. 58–60

Ask Nature

What Is Science? ما هو العلم؟

SCAN FOR PRACTICE

MUST KNOW



1. Latin **scientia** = knowledge; a **process based on inquiry** that explains events in nature
معرفة
2. Nature follows **rules**; scientists ask questions and observe to learn them
الطبيعة لها قواعد
3. Investigations: **observing** · **experiments with a control** · **modeling**
أنواع التحقيقات
4. **Life** science: living things
علوم الحياة
5. **Earth** science: Earth and space
علوم الأرض
6. **Physical** science: matter and energy · studies can overlap (artificial limbs)
علوم فيزيائية
7. **Thomson 1904**: electrons embedded in a ball of positive charge
طومسون
8. **Rutherford 1911**: nucleus with electrons orbiting like planets
رذرفورد
9. **Present day**: nucleus (protons + neutrons) + **electron cloud**; science stays open to change
السحابة الإلكترونية

Model · النموذج	Year · السنة	Idea · الفكرة
Thomson · طومسون	1904	electrons in a positive ball
Rutherford · رذرفورد	1911	nucleus + orbiting electrons
Present day · الحالي	today	p^+ + n nucleus + electron cloud

DON'T FALL FOR IT

✕ Science is just a list of facts.	✓ Science is a process of inquiry.	It explains events in nature. العلم عملية.
✕ Scientific ideas never change.	✓ They are modified with new evidence.	The atom model changed three times. العلم يتغير.
✕ Rutherford's model had an electron cloud.	✓ Rutherford: electrons orbit like planets.	The cloud is the present-day model. السحابة حديثة.

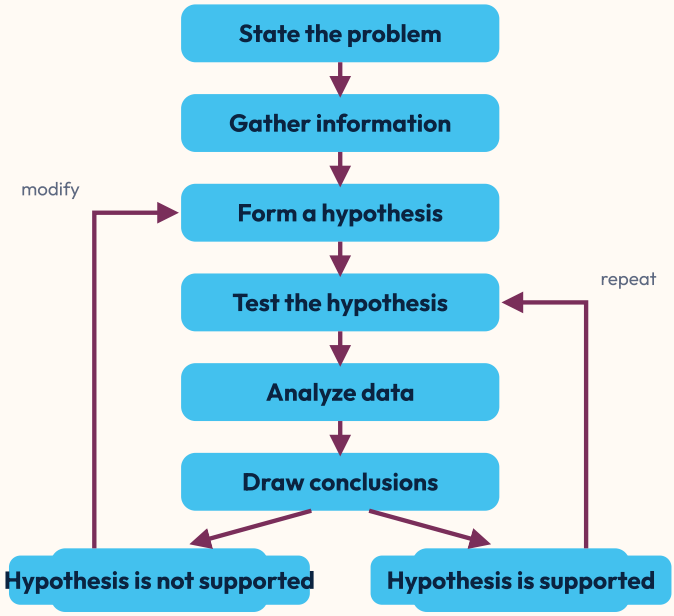
MODULE 1 · LESSON 1 · PART 1.2 · BOOK PP. 60–62

Test It

Scientific Methods الطرق العلمية

SCAN FOR PRACTICE

MUST KNOW



1. 1 **State the problem** (why/how) · 2 **Gather information** · 3 **Form a hypothesis**
   المشكلة · المعلومات · الفرضية
2. 4 **Test the hypothesis** · 5 **Analyze data** · 6 **Draw conclusions**
   اختبار · تحليل · استنتاج
3. Not rigid: add, repeat or skip steps; not supported → **modify** the hypothesis
   مرنة
4. **Hypothesis** = possible answer/solution based on what you know and observe (NASA ceramic idea)
   إجابة محتملة
5. **Experiment** = tests the effect of one thing on another using a **control**
   تجربة
6. Record ALL results, even unexpected; organize in tables and graphs; repeat to get the same data
   سجلي كل النتائج
7. **Independent**: what you change (microgravity) · **Dependent**: what changes/measured (plant growth)
   مستقل · تابع
8. **Constant**: factor that does not change (soil, plant type, light, water, temperature)
   ثابت
9. **Control**: standard for comparison (same plants grown on Earth)
   الضابط

Term · المصطلح	Meaning · المعنى	ISS example · مثال
Independent · مستقل	changed	microgravity
Dependent · تابع	measured	plant growth
Constant · ثابت	kept the same	soil, light, water
Control · ضابط	for comparison	plants on Earth

DON'T FALL FOR IT

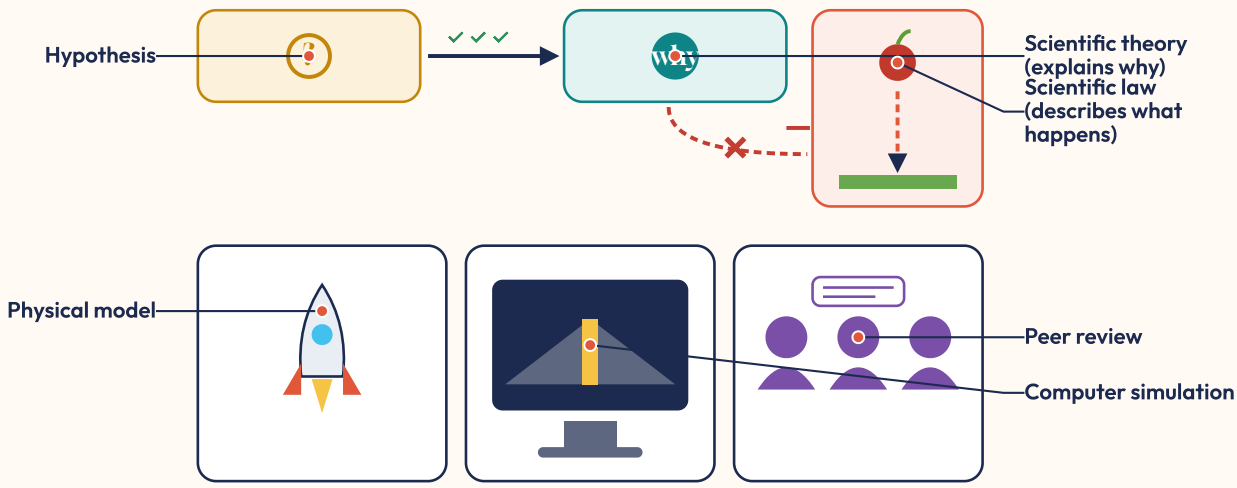
✕ The independent variable is measured.	✓ The DEPENDENT variable is measured.	Independent = what you change. التابع تقاس.
✕ Constant = control.	✓ Constant: kept the same; control: comparison group.	Different jobs. مختلفة.
✕ Throw away unexpected results.	✓ Record ALL results.	Discoveries come from surprises. سجل كل شيء.

Theory vs Law

Peer Review, Models, Theories and Laws مراجعة الأقران والنماذج والنظريات والقوانين

SCAN FOR PRACTICE

MUST KNOW



1. **Peer review**: scientists in the same field evaluate procedures and results
   مراجعة الأقران
2. **Bias**: expectations change analysis or conclusions; reduce with many trials, accurate notes, **blind experiments**
   التحيز
3. Valid experiments: **measurable** data and **repeatable**
   قابلة للقياس والتكرار
4. **Model** represents an idea, event or object; used when too large, small, slow or hazardous
   النموذج
5. History: Lord Kelvin's balls in jelly (light through space); atom models
   كيلفن
6. **Computer simulation**: safely mimics a process and collects data (flight simulators)
   المحاكاة
7. **Theory**: explanation from many observations and investigations; not a guess; can be modified
   نظرية = لماذا
8. **Law**: describes what happens, seems true all the time; does not explain why (gravity)
   قانون = ماذا
9. Theories do **not** become laws · science can't answer opinions or values
   النظرية لا تصبح قانونًا

	Theory · نظرية	Law · قانون
Job · الوظيفة	explains WHY / HOW	describes WHAT happens
Based on · مبني على	many investigations	observations always true
Can change? · تتغير؟	yes, can be modified	describes specific relationships
Becomes the other? · تتحول؟	never becomes a law	can be explained by a theory

DON'T FALL FOR IT

✕ A theory becomes a law with enough evidence.	✓ Theories never become laws.	They do different jobs. النظرية لا تصبح قانونًا.
✕ A theory is a guess.	✓ It is based on many investigations.	A guess is not a theory. النظرية ليست تخمينًا.
✕ A law explains why.	✓ A law describes what happens.	The theory explains why. القانون يصف.

LAST LOOK · MODULE 1 · LESSON 1 · THE NIGHT BEFORE THE EXAM

The Methods of Science

Book pp. 58–65 طرق العلم

LESSON TEST

MIND MAP · THE WHOLE LESSON

1.1 · Ask Nature — ما هو العلم؟

- Latin scientia = knowledge; a process based on inquiry that explains events in nature
- Nature follows rules; scientists ask questions and observe to learn them
- Investigations: observing · experiments with a control · modeling
- Life science: living things

1.2 · Test It — الطرق العلمية

- 1 State the problem (why/how) · 2 Gather information · 3 Form a hypothesis
- 4 Test the hypothesis · 5 Analyze data · 6 Draw conclusions
- Not rigid: add, repeat or skip steps; not supported → modify the hypothesis
- Hypothesis = possible answer/solution based on what you know and observe (NASA ceramic idea)

The Methods of Science طرق العلم

1.3 · Theory vs Law — مراجعة الأقران والنماذج وا

- Peer review: scientists in the same field evaluate procedures and results
- Bias: expectations change analysis or conclusions; reduce with many trials, accurate notes, blind experiments
- Valid experiments: measurable data and repeatable
- Model represents an idea, event or object; used when too large, small, slow or hazardous

MEMORY HOOKS

- **L · E · P** · Life, Earth, Physical: the three branches — الفروع الثلاثة
- **I change, it Depends** · Independent = I change it · Dependent = depends on it — أنا بغيّر
- **Law = What · Theory = Why** · Laws describe, theories explain — ماذا ولماذا
- **Pudding → planets → cloud** · Thomson, Rutherford, present day — تطور نموذج الذرة
- **Smart Girls Find The Answer Daily** · State, Gather, Form, Test, Analyze, Draw — ترتيب الخطوات
- **Blind = no peeking** · Researchers don't know which group is which — بدون معرفة

WORDS TO KNOW

Science العلم	**Life science** علوم الحياة	**Earth science** علوم الأرض	**Physical science** العلوم الفيزيائية
Electron cloud السحابة الإلكترونية	**Investigation** التحقيق العلمي	**Scientific methods** الطرق العلمية	**Hypothesis** الفرضية
Experiment التجربة	**Variable** المتغير	**Independent variable** المتغير المستقل	**Dependent variable** المتغير التابع
Constant الثابت	**Control** الضابط (المجموعة الضابطة)	**Peer review** مراجعة الأقران	**Bias** التحيز
Blind experiment التجربة العمياء	**Model** النموذج	**Computer simulation** المحاكاة الحاسوبية	**Scientific theory** النظرية العلمية
Scientific law القانون العلمي			

In the Catalyst mock exam: Q19, Q20, Q21, Q30 · scan the QR for this lesson's full test